ProteinNetworkSight Tutorial

Tutorial PDF

Summary

ProteinNetworkSight is a web-based tool for constructing patient-specific protein interaction networks from user-provided gene or protein scores. The platform integrates quantitative input data with interaction information retrieved from the STRING database to identify biologically relevant protein clusters. The resulting networks are further linked to the Cancer Drugs Database, enabling the exploration of potential therapeutic targets.

If you use ProteinNetworkSight, please cite:
Nahor O, Migdal N, Gibli A, Tsvitman T, Eldad A, Raveh S, Polinovski G, Zaid D, Kravchenko-Balasha N, Cohen NE. ProteinNetworkSight: A User-Friendly Platform for Transforming Co-Expression Patterns into Actionable Therapeutic Insights Through Interactive Network Visualization. Nucleic Acids Research. 2026.

* ProteinNetworkSight doesnt track user data or activity, requires no login and is available for free use.

Open source code: Active Github Repository, Initial release

ProteinNetworkSight integrates information from three external databases:

a. STRING-db - a database of known and predicted protein-protein interactions.
b. Cancer Drugs Database - a resource containing licensed anticancer drugs.
c. UniProt - a protein sequence and functional information resource.

The generated networks allow interactive exploration of protein interactions and facilitate the identification of biologically important proteins and potential drug targets.

Workflow Overview

The ProteinNetworkSight analysis consists of three main steps:

1. Upload input data - Users upload a tabular file containing protein or gene identifiers together with numeric scores (e.g., PCA loadings, fold changes, or other quantitative metrics).
2. Define analysis parameters - Users specify the identifier column, numeric column prefix, interaction score thresholds, and organism of interest.
3. Explore the resulting networks - The platform constructs a protein interaction network using interaction data from the STRING database. Users can interactively explore the network, adjust visualization settings, and export the results in graphical or tabular formats.

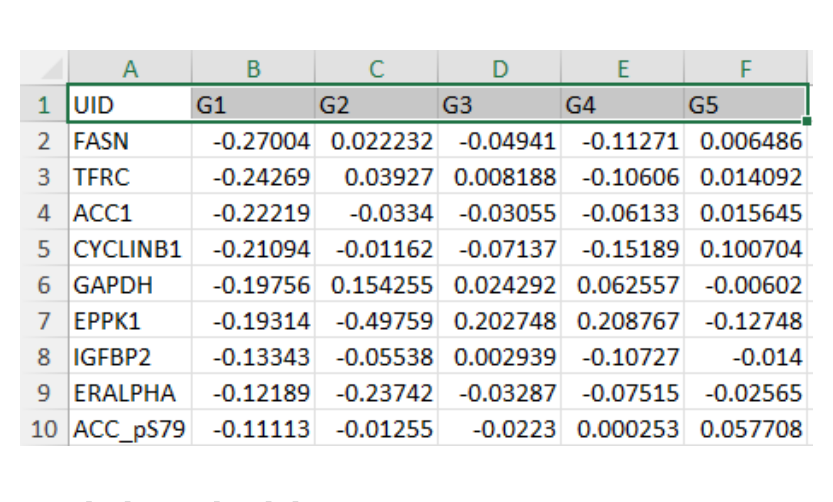
Starting Point

To begin the analysis, users must upload an Excel, CSV or a TSV containing a single sheet of tabular data. Each row should correspond to a protein or gene, and each column should represent a feature. The file must contain no more than 5000 rows. One column should contain protein or gene identifiers, while additional columns should contain numeric values representing gene/protein scores or quantitative metrics.

The first row must contain column headers. Numeric columns intended for analysis must share a common prefix.

An example file is provided (adapted from Vasudevan et al., npj Precision Oncology, 2021, Supplementary Data 1). In this example, the gene identifier column is labeled “UID”, and the numeric columns begin with the prefix “G”, followed by an index.

In this dataset, the G columns represent protein weights derived from the information-theoretic analysis described in Vasudevan et al. However, the format is flexible, and these columns may contain any quantitative scores, coefficients, or fold-change values, depending on the method used to identify co-expression patterns.

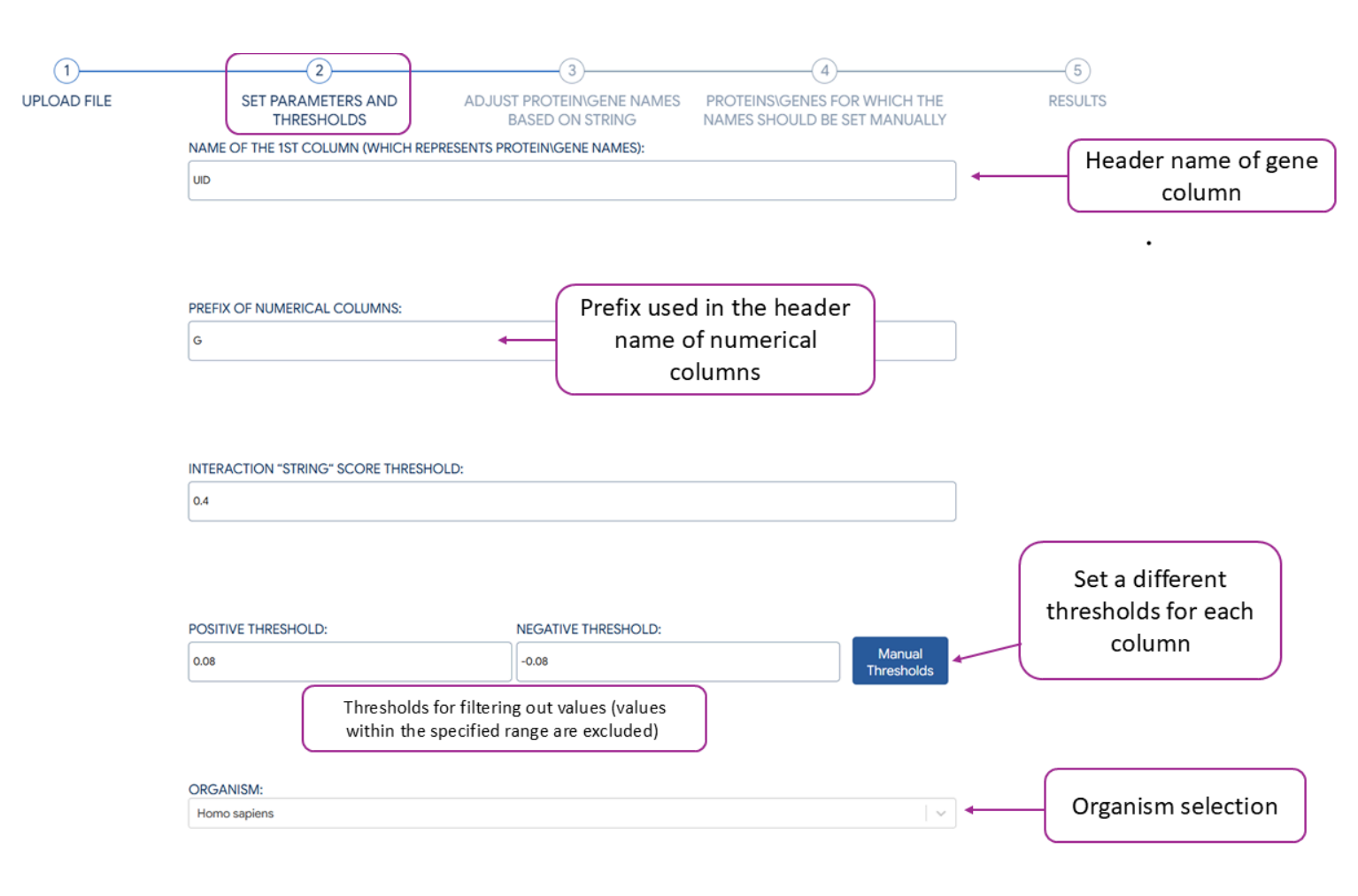


	A	B	C	D	E	F
1	UID	G1	G2	G3	G4	G5
2	FASN	-0.27004	0.022232	-0.04941	-0.11271	0.006486
3	TFRC	-0.24269	0.03927	0.008188	-0.10606	0.014092
4	ACC1	-0.22219	-0.0334	-0.03055	-0.06133	0.015645
5	CYCLINB1	-0.21094	-0.01162	-0.07137	-0.15189	0.100704
6	GAPDH	-0.19756	0.154255	0.024292	0.062557	-0.00602
7	EPPK1	-0.19314	-0.49759	0.202748	0.208767	-0.12748
8	IGFBP2	-0.13343	-0.05538	0.002939	-0.10727	-0.014
9	ERALPHA	-0.12189	-0.23742	-0.03287	-0.07515	-0.02565
10	ACC_pS79	-0.11113	-0.01255	-0.0223	0.000253	0.057708

Setting parameters and thresholds

In this step the user should specify:

a. Identifier column - The header name of the protein/gene identifier column (e.g., “UID” in the example file).
b. Numeric column prefix - The prefix shared by the numeric columns (e.g., “G”, where Gi represents the weight of each protein in patterns such as G1, G2, G3).
c. STRING interaction threshold - The lower limit for the STRING interaction score, representing the probability of interaction between pairs of proteins.
d. Score thresholds - Positive and negative thresholds used to exclude proteins whose scores fall within a specified range.
e. Organism of interest -For example, Homo sapiens in the example dataset.


1 UPLOAD FILE
2 SET PARAMETERS AND THRESHOLDS
3 ADJUST PROTEIN/GENE NAMES BASED ON STRING
4 PROTEINS/GENES FOR WHICH THE NAMES SHOULD BE SET MANUALLY
5 RESULTS
Manual Thresholds

NAME OF THE 1ST COLUMN (WHICH REPRESENTS PROTEIN/GENE NAMES): UID
PREFIX OF NUMERICAL COLUMNS: G
INTERACTION “STRING” SCORE THRESHOLD: 0.4
POSITIVE THRESHOLD: 0.08
NEGATIVE THRESHOLD: -0.08
ORGANISM: Homo sapiens

Header name of gene column

Prefix used in the header name of numerical columns

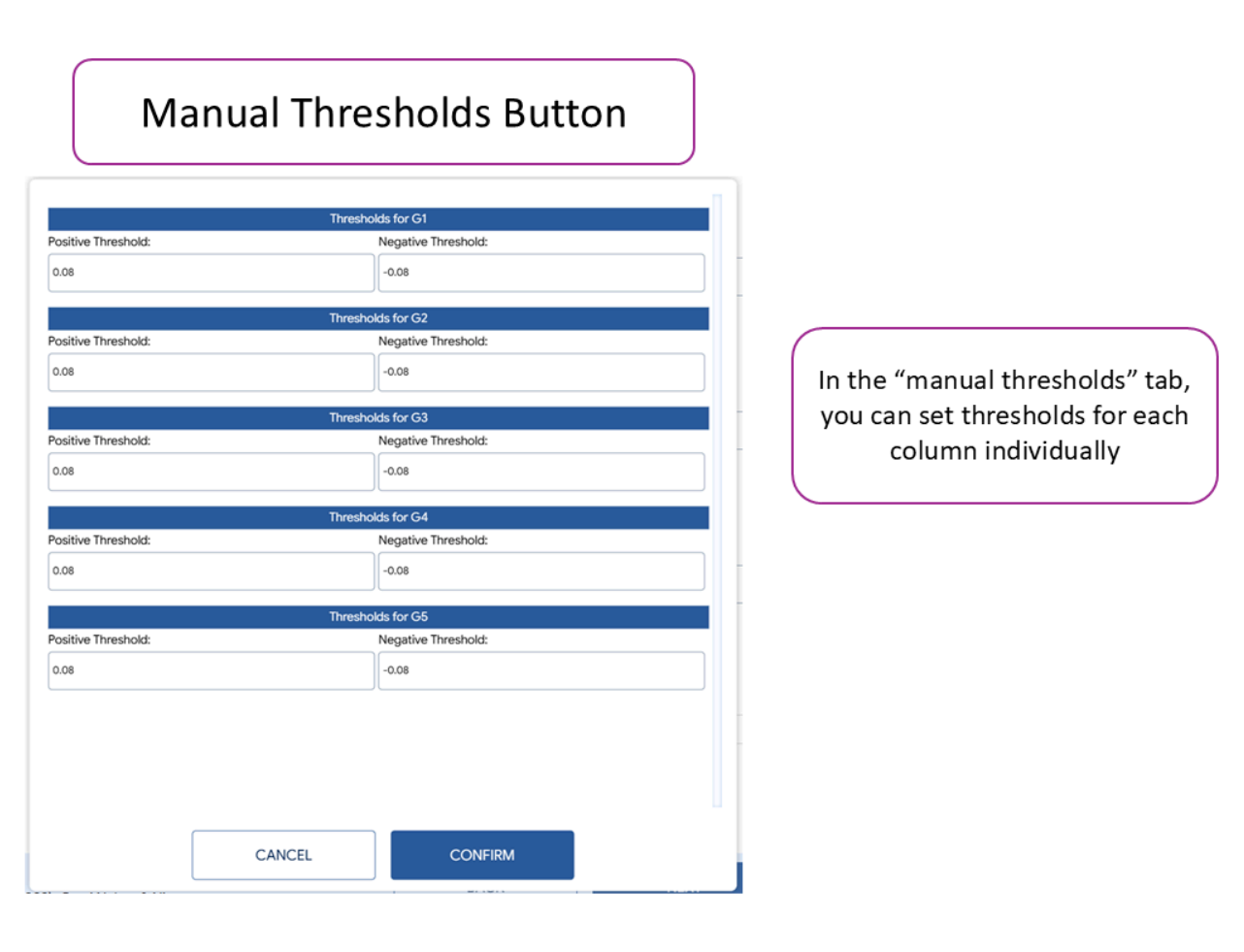
Set a different thresholds for each column

Thresholds for filtering out values (values within the specified range are excluded)

Organism selection

Manual thresholds adjustments

- You can also choose to adjust the thresholds of every score (G in the example) column in the graph individually.

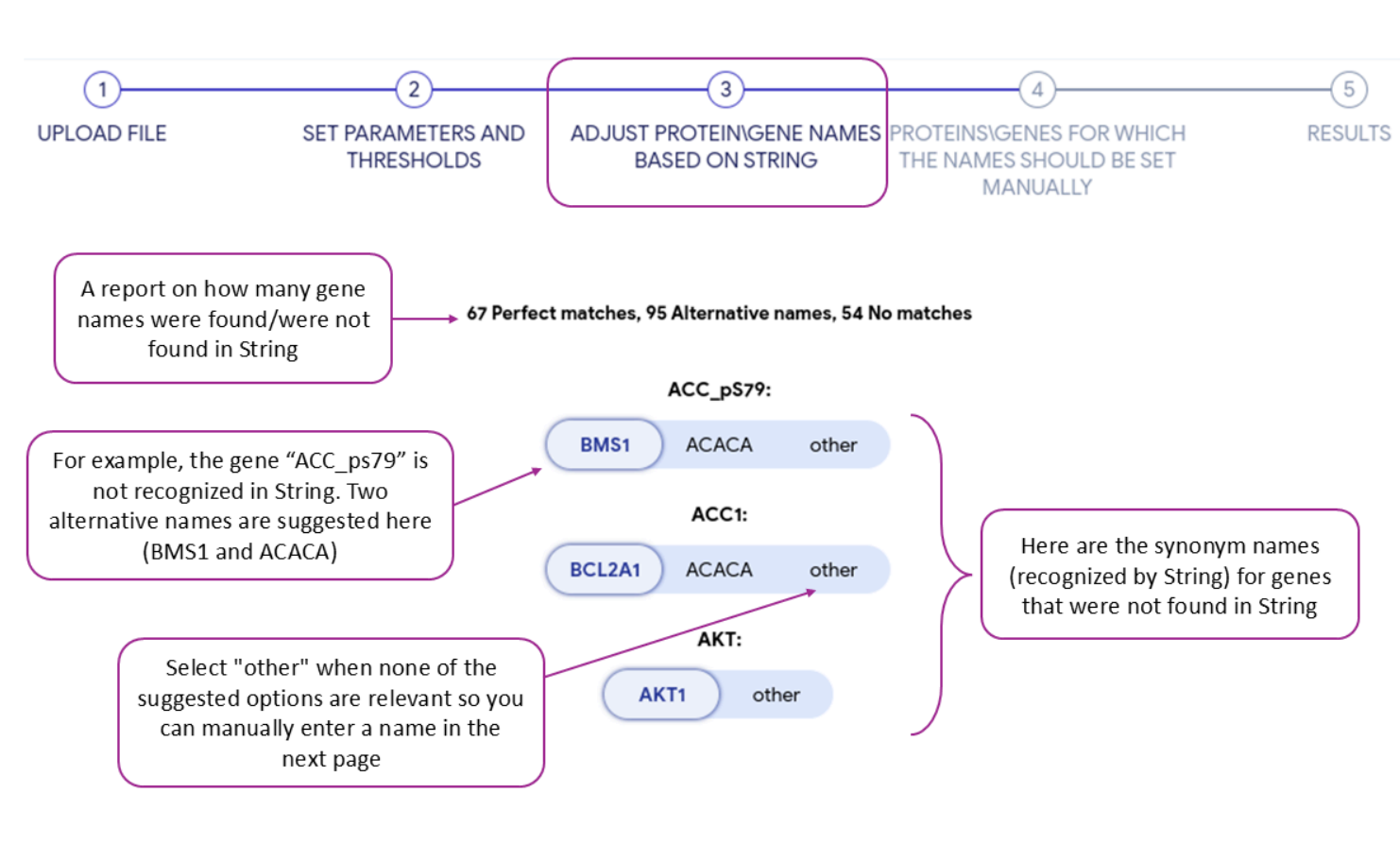

Manual Thresholds Button
Thresholds for G1
Thresholds for G2
Thresholds for G3
Thresholds for G4
Thresholds for G5
CANCEL
CONFIRM

Positive Threshold: 0.08 Negative Threshold: -0.08

In the “manual thresholds” tab, you can set thresholds for each column individually

Adjusting protein/gene names

ProteinNetworkSight retrieves interaction data from STRING-db in order to construct and visualize protein interaction networks. Therefore, protein or gene identifiers in the input file should match the names used in STRING-db. Because genes often have multiple synonyms, some identifiers in the input file may not match automatically. The following steps highlight such cases and allow users to select alternative identifiers.
in step 3, The system suggests alternative identifiers recognized by the STRING database. For each gene, users can choose one of the suggested identifiers or manually enter an alternative identifier.


1 UPLOAD FILE
2 SET PARAMETERS AND THRESHOLDS
3 ADJUST PROTEIN/GENE NAMES BASED ON STRING
4 PROTEINS/GENES FOR WHICH THE NAMES SHOULD BE SET MANUALLY
5 RESULTS
ACC_pS79:
BMS1 ACACA other
ACC1:
BCL2A1 ACACA other
AKT:
AKT1 other

67 Perfect matches, 95 Alternative names, 54 No matches

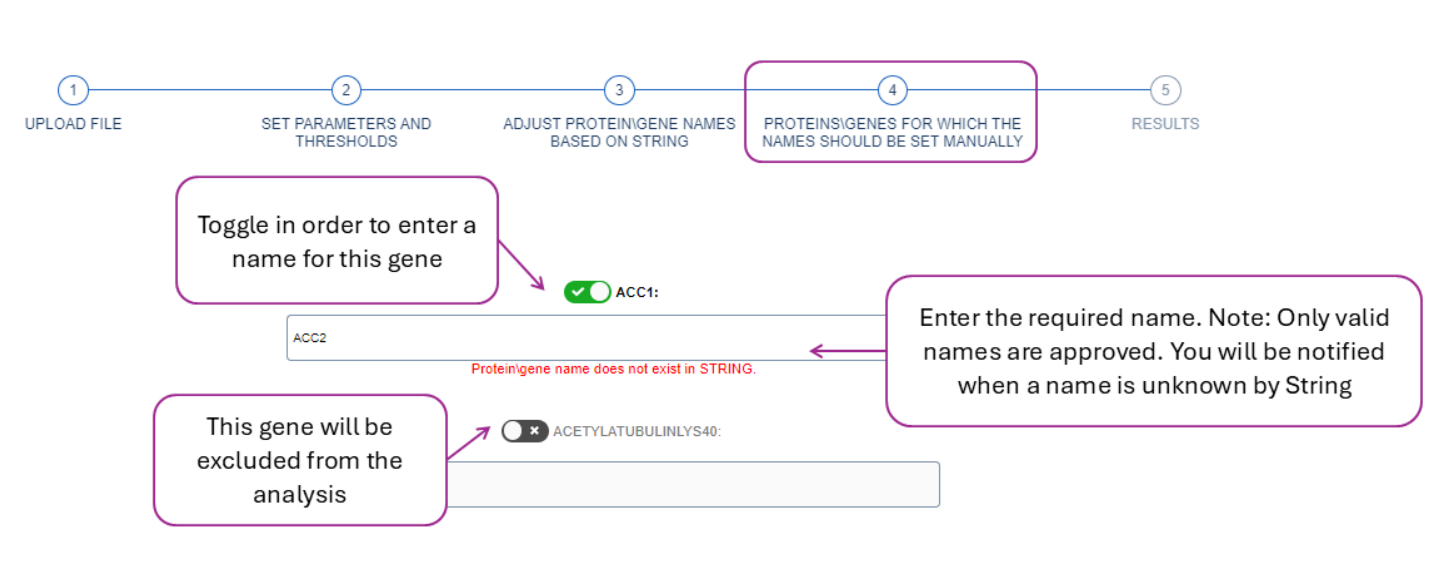
A report on how many gene names were found/were not found in String

For example, the gene “ACC_ps79” is not recognized in String. Two alternative names are suggested here (BMS1 and ACACA)

Select “other” when none of the suggested options are relevant so you can manually enter a name in the next page

Here are the synonym names (recognized by String) for genes that were not found in String

For genes for which no alternative identifier was found automatically, in step 4 users may manually enter a valid identifier. Genes that remain unmatched will be excluded from the network analysis.


1 UPLOAD FILE
2 SET PARAMETERS AND THRESHOLDS
3 ADJUST PROTEIN/GENE NAMES BASED ON STRING
4 PROTEINS/GENES FOR WHICH THE NAMES SHOULD BE SET MANUALLY
5 RESULTS
ACC1:
ACC2
ACETYLATUBULINLYS40

Protein/gene name does not exist in STRING.

Toggle in order to enter a name for this gene

Enter the required name. Note: Only valid names are approved. You will be notified when a name is unknown by String

This gene will be excluded from the analysis

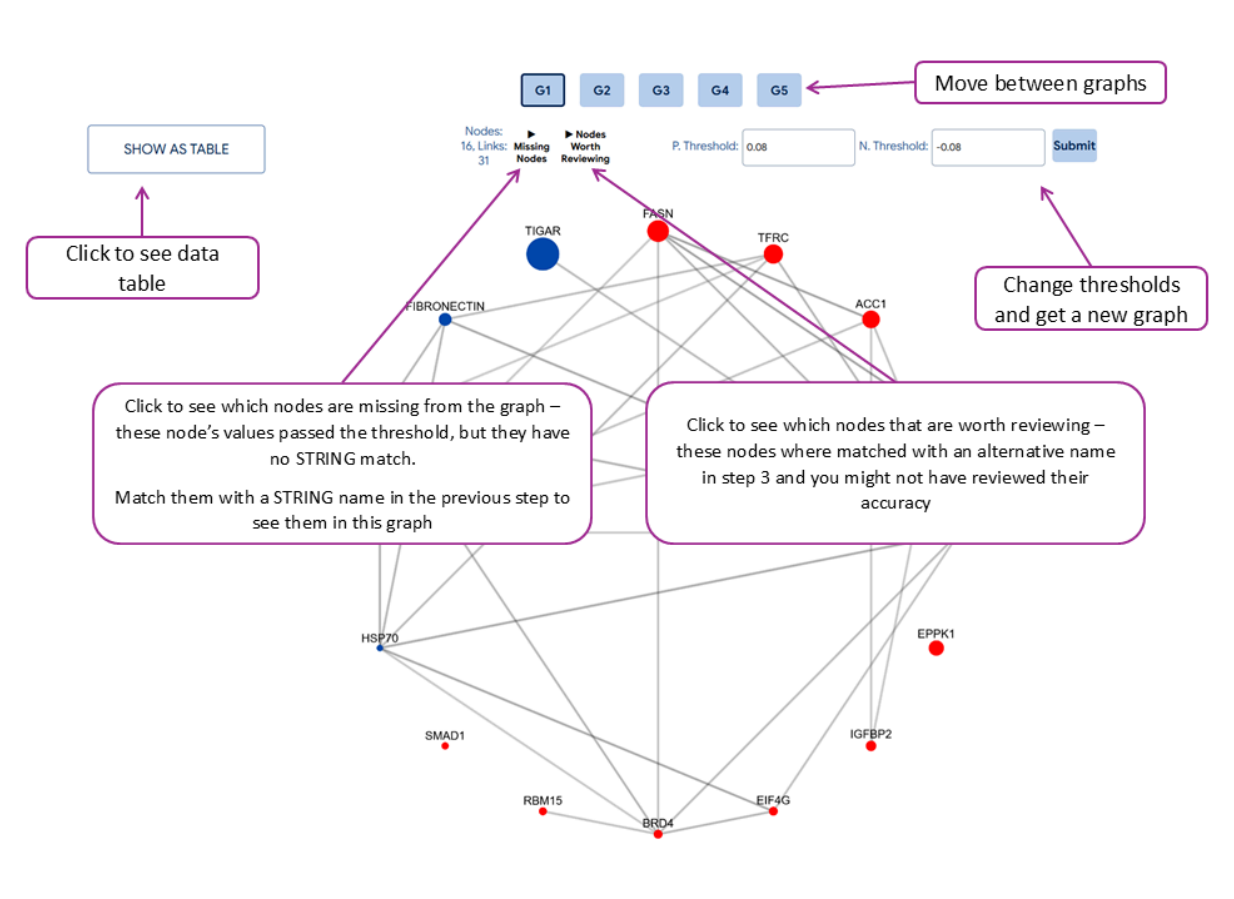
Results page

The resulting network contains nodes representing proteins, where the node size is proportional to the input protein score. Edges are derived from STRING, and edge width represents the probability of a protein-protein interaction. Results can be viewed and downloaded in either a tabular or network format. The network view provides an interactive visualization, allowing users to zoom, drag nodes, and explore the network structure.

Graph representation

The graphical representation displays a network in which each node represents a protein.

- Node size is proportional to the input score value.
- Node color indicates the sign of the value (by default blue for positive values and red for negative values).
- Edges represent functional protein-protein interactions, with edge width corresponding to the interaction probability.


G1 G2 G3 G4 G5
SHOW AS TABLE
Missing Nodes
Nodes worth Reviewing
P. Threshold: 0.08
N. Threshold: -0.08
Submit
TIGAR
FASN
TFRC
FIBRONECTIN
ACC1
HSP70
EPPK1
IGFBP2
SMAD1
RBM15
BRD4
EIF4G

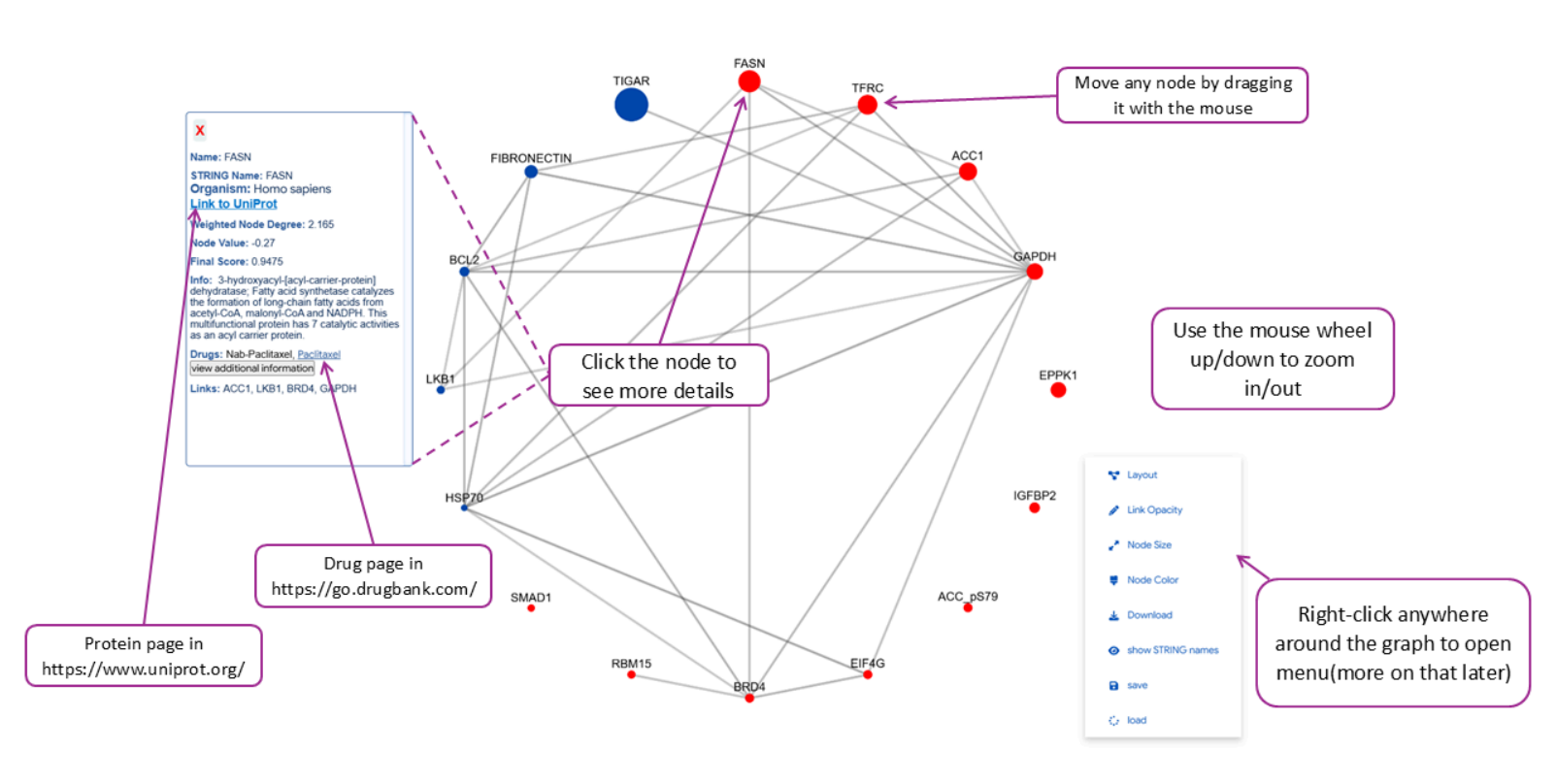
Move between graphs

Click to see data table

Change thresholds and get a new graph

Click to see which nodes are missing from the graph – these node's values passed the threshold, but they have no STRING match.
Match them with a STRING name in the previous step to see them in this graph

Click to see which nodes that are worth reviewing – these nodes where matched with an alternative name in step 3 and you might not have reviewed their accuracy


TIGAR
FASN
TFRC
FIBRONECTIN
ACC1
BCL2
LKB1
GAPDH
HSP70
EPPK1
IGFBP2
SMAD1
ACC_pS79
RBM15
BRD4
EIF4G
Layout
Link Opacity
Node Size
Node Color
Download
show STRING names
save
load

Weighted Node Degree: 2.165
Node Value: -0.27
Final Score: 0.9475
Drugs: Nab-Paclitaxel, Paclitaxel
Links: ACC1, LKB1, BRD4, GAPDH

Move any node by dragging it with the mouse

Click the node to see more details

Use the mouse wheel up/down to zoom in/out

Drug page in https://go.drugbank.com/

Protein page in https://www.uniprot.org/

Right-click anywhere around the graph to open menu(more on that later)

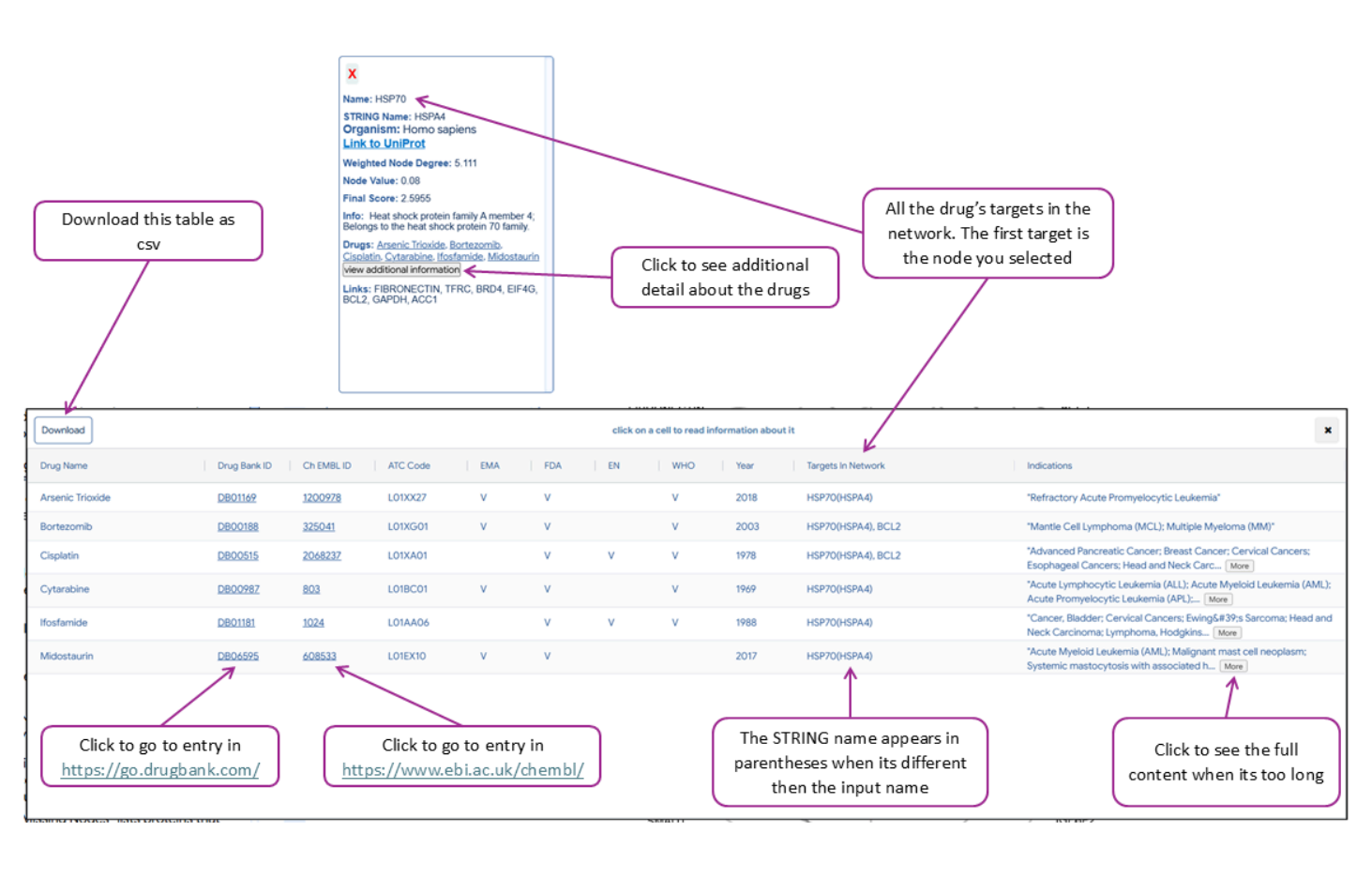
In the illustrated example, multiple protein interactions are shown. For example:,

- TIGAR appears with a large positive value (blue node) and interacts with GAPDH, which has a smaller negative value.
- ACC1 interacts with FASN, LKB1 and GAPDH, with the thicker edge indicating a stronger interaction probability with FASN.

Clicking on a node reveals additional information about the corresponding protein.
Users can also drag nodes interactively to improve the layout and facilitate interpretation of the network.

For the user's convinience, we provide 2 collapsable lists to help the user validate the accuracy of the network.

1. Missing Nodes - A lists of nodes that could be included in the graph if the user matched them to a valid identifier in the previous step.
2. Nodes worth reviewing - A list of nodes in the graph that the user might have missed validating their accuracy, because they were not inserted manually


Download
click on a cell to read information about it

Name: HSP70
STRING Name: HSPA4
Organism: Homo sapiens
Link to UniProt
Weighted Node Degree: 5.111
Node Value: 0.08
Final Score: 2.5955
Info: Heat shock protein family A member 4; Belongs to the heat shock protein 70 family.
Drugs: Arsenic Trioxide, Bortezomib, Cisplatin, Cytarabine, Ifosfamide, Midostaurin
view additional information
Links: FIBRONECTIN, TFRC, BRD4, EIF4G, BCL2, GAPDH, ACC1

Drug Name	Drug Bank ID	Ch EMBL ID	ATC Code	EMA	FDA	EN	WHO	Year	Targets in Network	Indications
Arsenic Trioxide	DB01169	1200978	L01XX27	V	V		V	2018	HSP70(HSPA4)	"Refractory Acute Promyelocytic Leukemia"
Bortezomib	DB00188	325041	L01XG01	V	V		V	2003	HSP70(HSPA4), BCL2	"Mantle Cell Lymphoma (MCL); Multiple Myeloma (MM)"
Cisplatin	DB00515	2068237	L01XA01		V	V	V	1978	HSP70(HSPA4), BCL2	"Advanced Pancreatic Cancer; Breast Cancer; Cervical Cancers; Esophageal Cancers; Head and Neck Carc... More
Cytarabine	DB00987	803	L01BC01	V	V		V	1969	HSP70(HSPA4)	"Acute Lymphocytic Leukemia (ALL); Acute Myeloid Leukemia (AML); Acute Promyelocytic Leukemia (APL);... More
Ifosfamide	DB01181	1024	L01AA06		V	V	V	1988	HSP70(HSPA4)	"Cancer; Bladder; Cervical Cancers; Ewing&#39;s Sarcoma; Head and Neck Carcinoma; Lymphoma, Hodgkins... More
Midostaurin	DB06595	608533	L01EX10	V	V			2017	HSP70(HSPA4)	"Acute Myeloid Leukemia (AML); Malignant mast cell neoplasm; Systemic mastocytosis with associated h... More

Download this table as csv

Click to see additional detail about the drugs

All the drug's targets in the network. The first target is the node you selected

Click to go to entry in https://go.drugbank.com/

Click to go to entry in https://www.ebi.ac.uk/chembl/

The STRING name appears in parentheses when its different then the input name

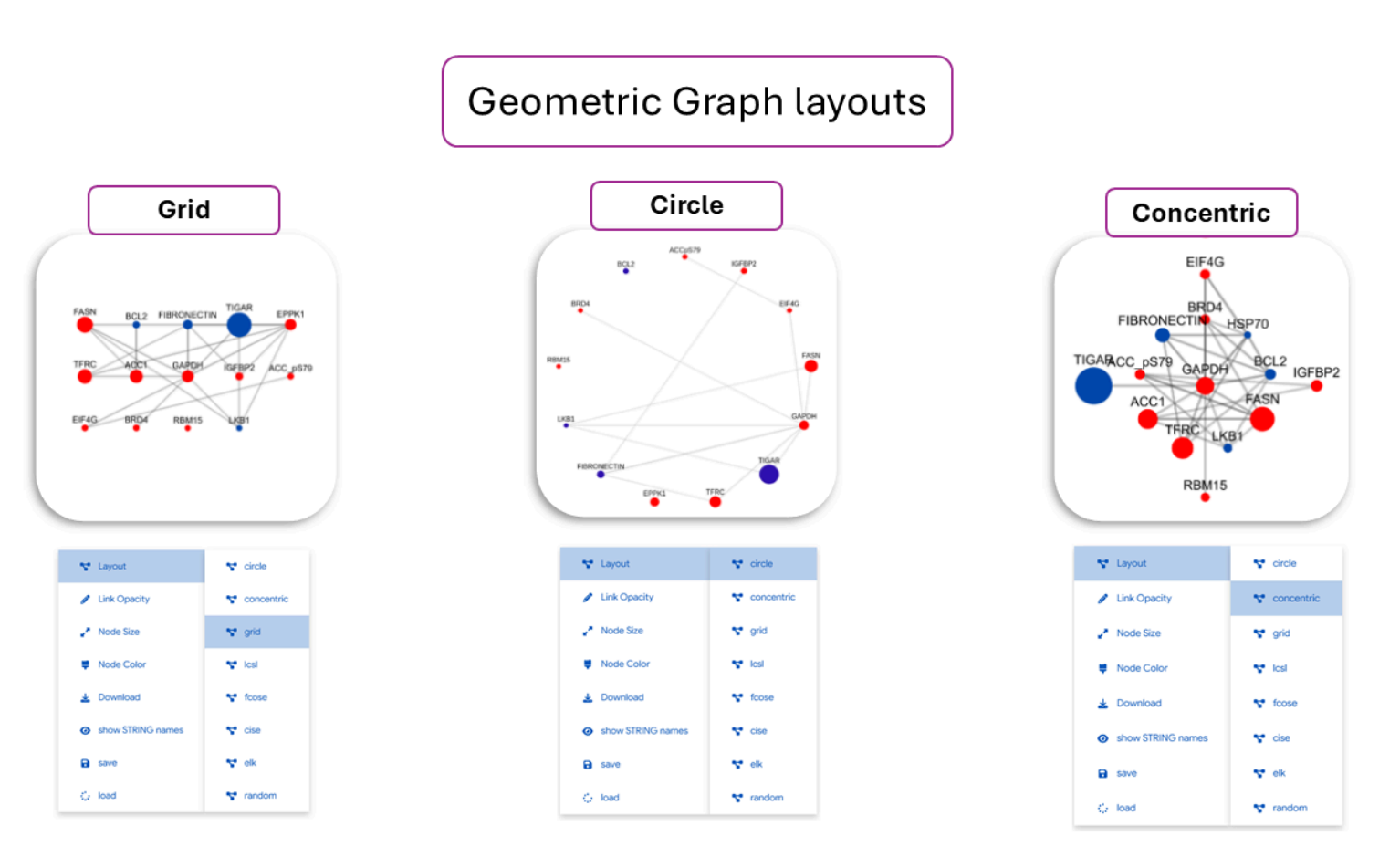
Click to see the full content when its too long

After identifying a key node of interest, open the full drug details list to explore potential drug combinations.

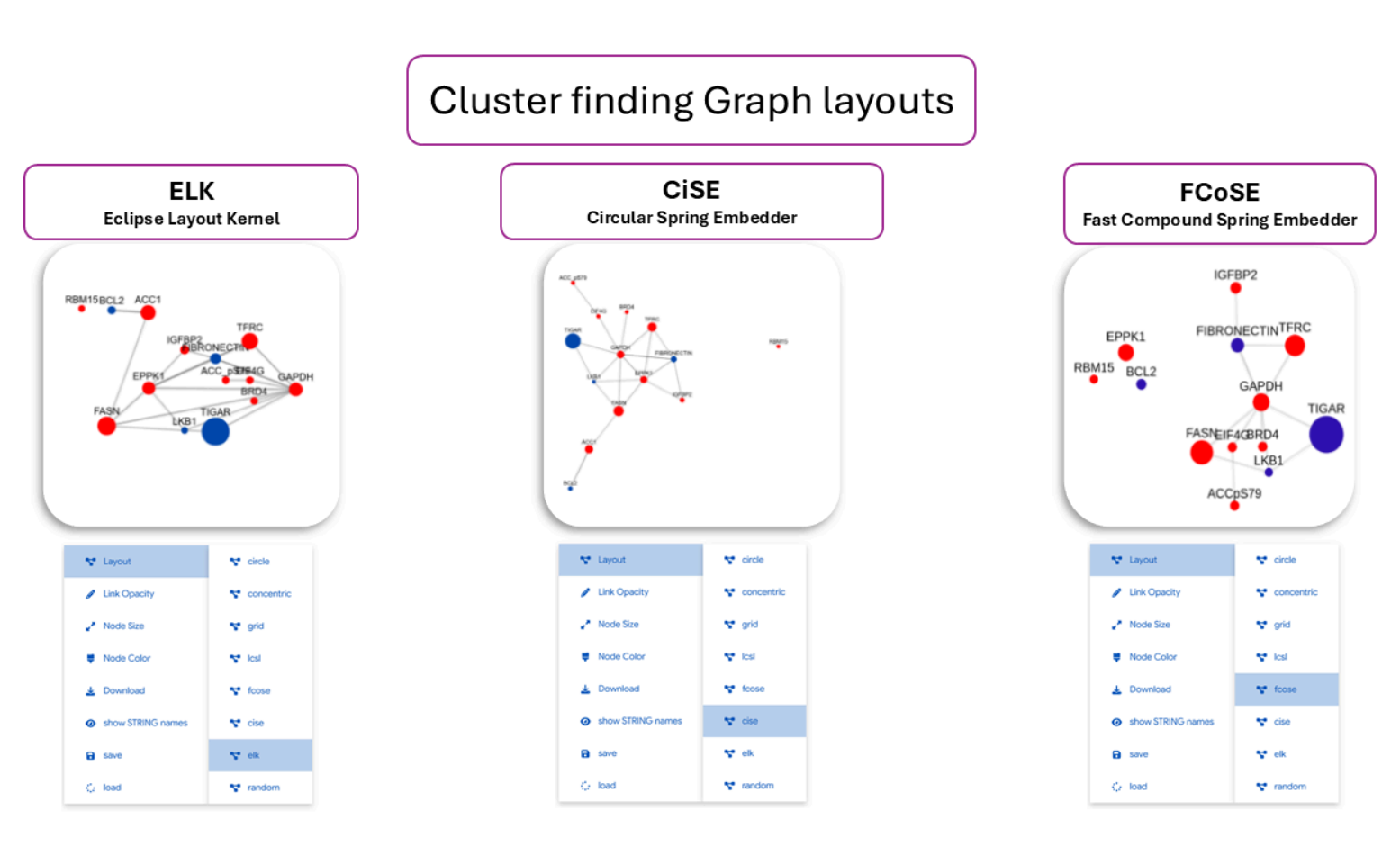
The right-click context menu in the graph visualization allows users to modify visual properties and customize the appearance of the network.

Multiple layout algorithms are available to organize the graph and reveal structural patterns.
These layouts help identify biologically relevant proteins in each network, that may guide the selection of effective drug combinations.

Presented below are three fundamental options that are broadly applicable to any graph.


Geometric Graph layouts
Grid
Circle
Concentric
Layout
Link Opacity
Node Size
Node Color
Download
show STRING names
save
load
circle
concentric
grid
lcsl
fcose
cise
elk
random

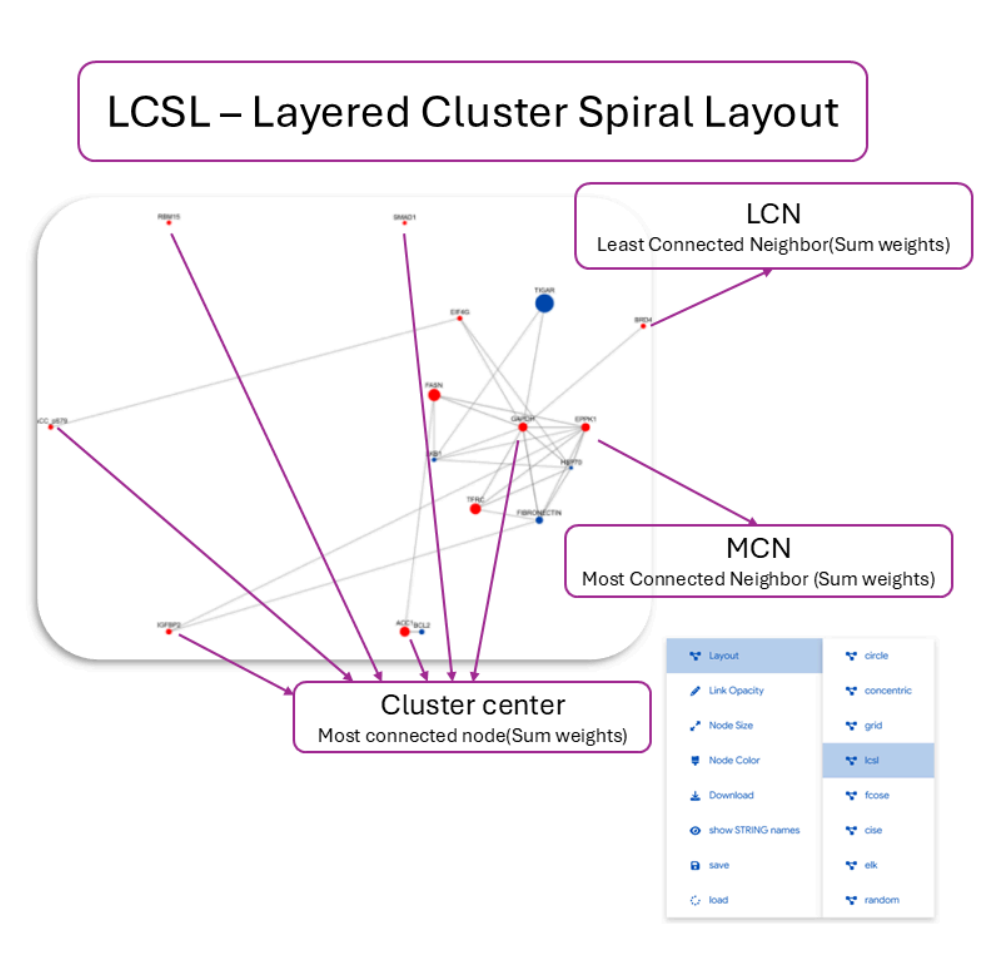
Additionally, advanced layout options utilizing various clustering algorithms are available to provide an alternative visualization of the graph's structure.
We recommend exploring each of these options, as they can uncover valuable insights and reveal unique properties of the graph.

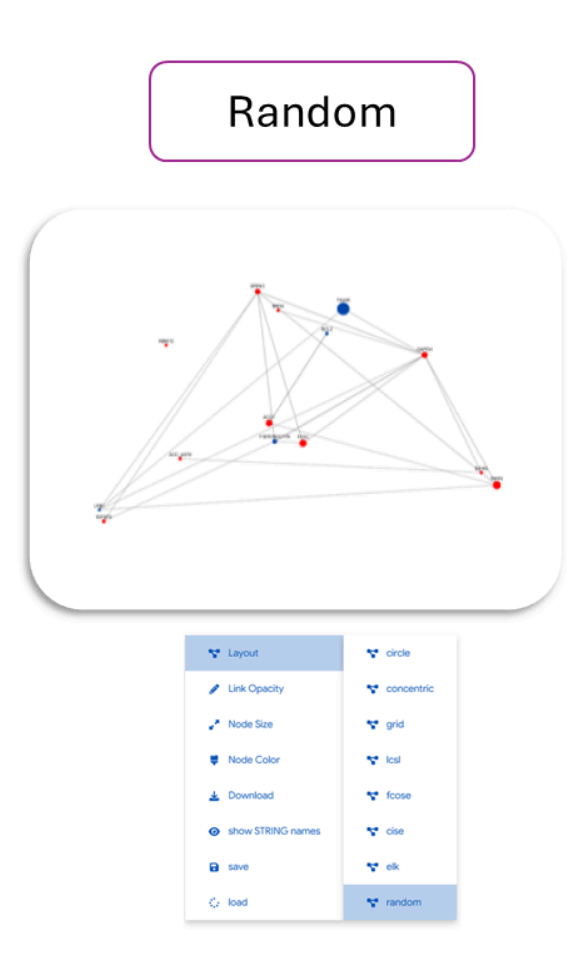

Cluster finding Graph layouts
ELK
Eclipse Layout Kernel
CiSE
Circular Spring Embedder
FCoSE
Fast Compound Spring Embedder

Further information about these layouts can be found in the cytoscape.js layout documentation.
If you would like us to add additional layouts, please contact us.

Additionally, for users who prefer a more complex approach, the tool also provides a custom clustering algorithm, called LCSL (Layered Cluster Spiral Layout). Unlike traditional layouts that rely mainly on link counts, LCSL incorporates link weights when identifying clusters. Nodes within each cluster are arranged in a spiral pattern based on the weighted connectivity of each node. This layout is designed to facilitate the identification of important proteins and their associated therapeutic targets.

For users who prefer a simpler apprach, we also provide a Random layout, which randomly places the nodes on the graph. You can click it multiple times to see different random results.

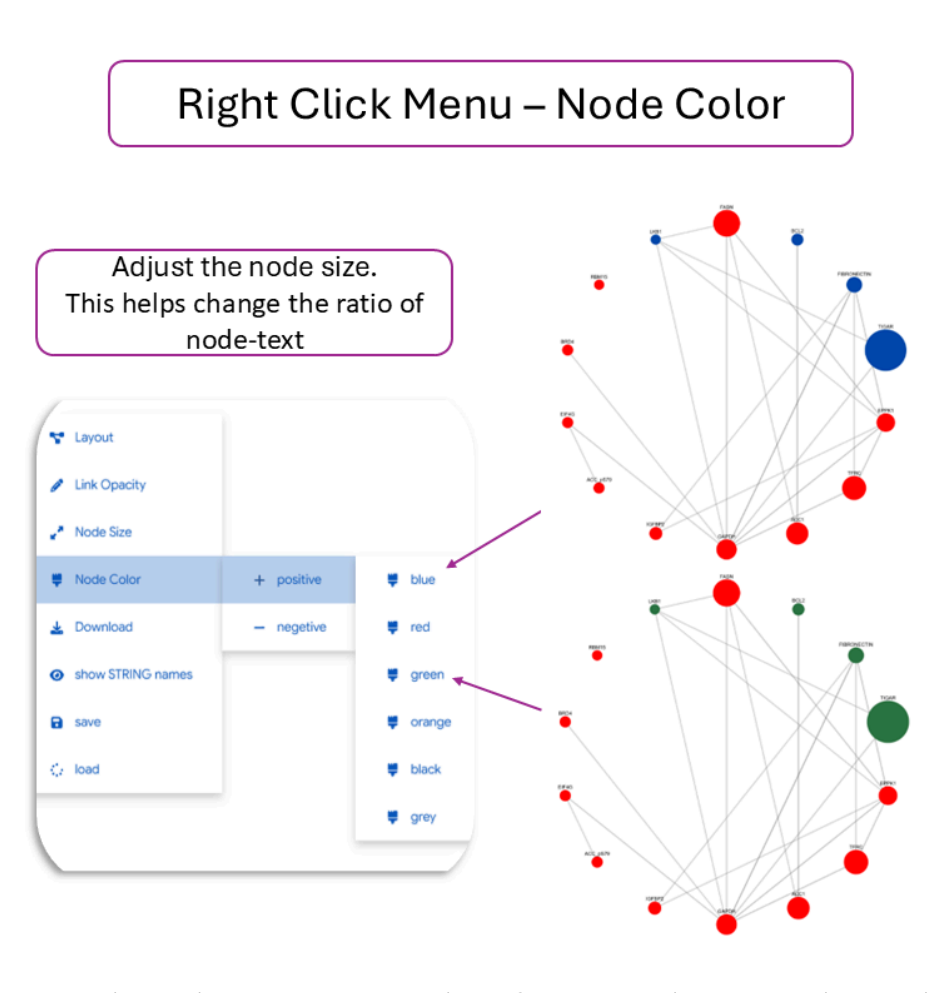

LCSL – Layered Cluster Spiral Layout
LCN
Least Connected Neighbor(Sum weights)
MCN
Most Connected Neighbor (Sum weights)
Cluster center
Most connected node(Sum weights)



Graph Appearance Options

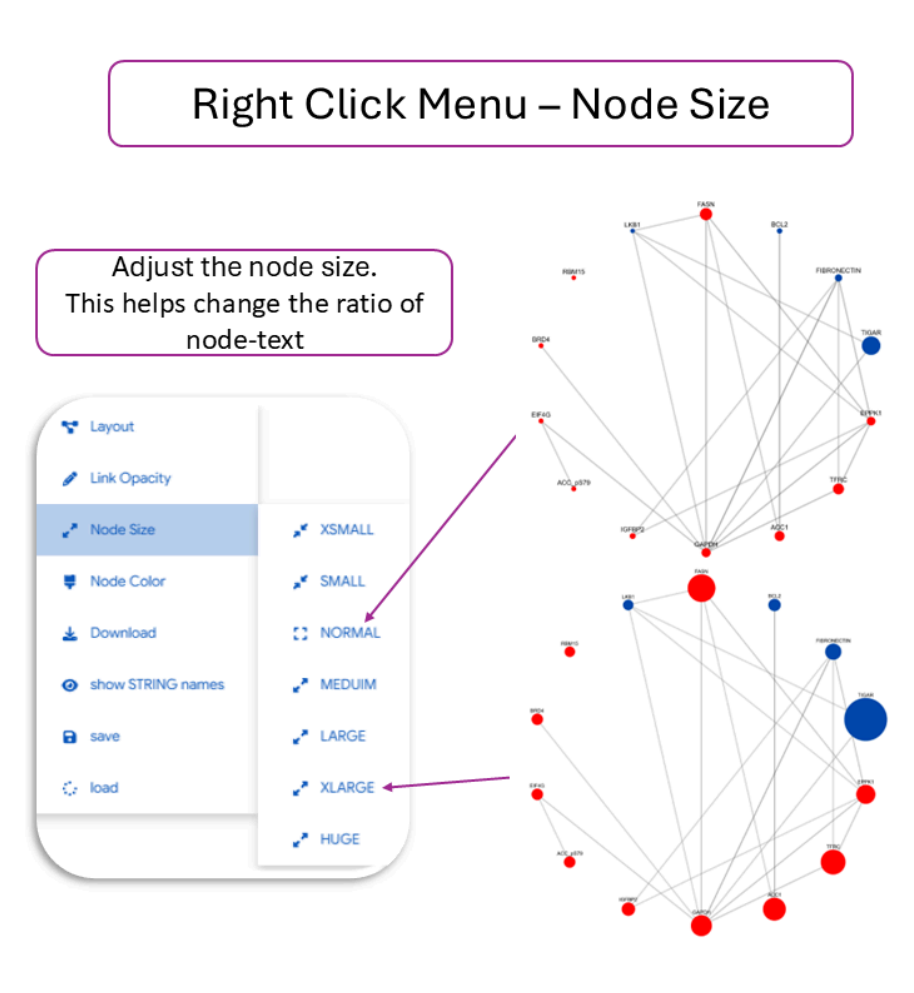
Several visual options are available to improve readability.

Node color allows selection of one of six predefined color schemes, applied separately to positive and negative nodes.


Right Click Menu – Node Color
positive
negative
blue
red
green
orange
black
grey

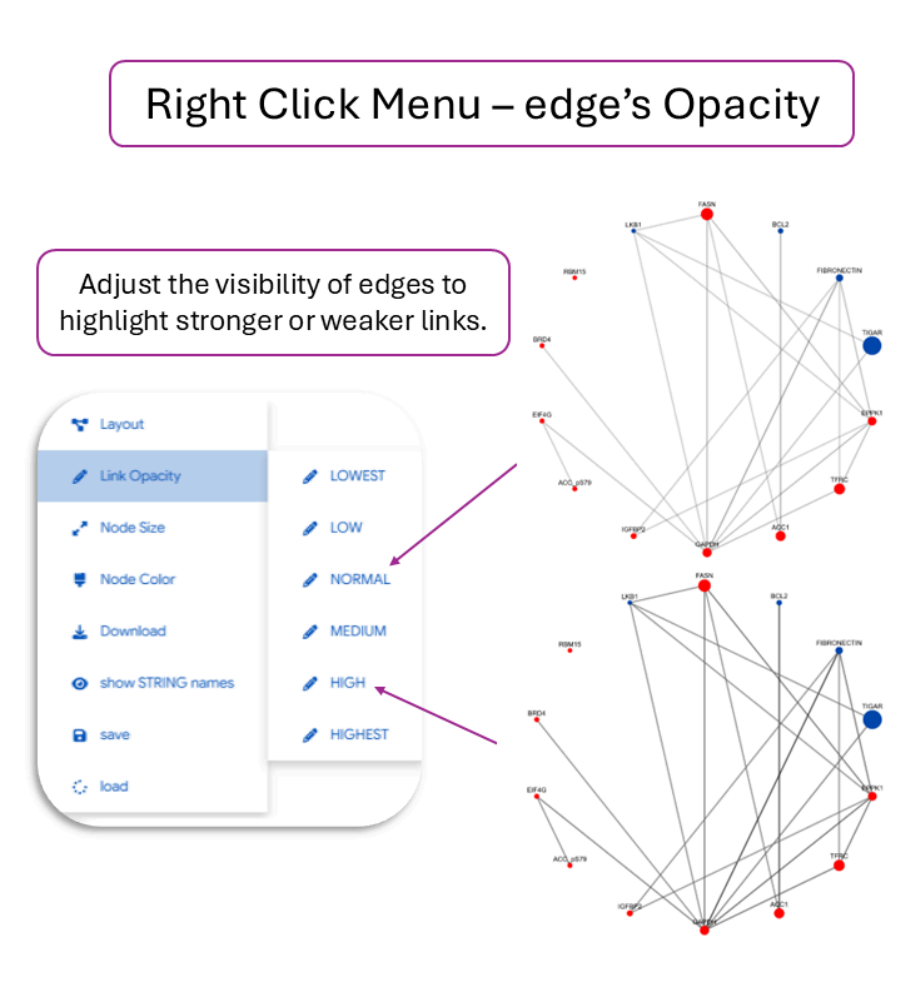
Adjust the node size. This helps change the ratio of node-text

Node size adjusts the overall size of nodes while preserving their relative size differences.


Right Click Menu – Node Size
XSMALL
SMALL
NORMAL
MEDIUM
LARGE
XLARGE
HUGE

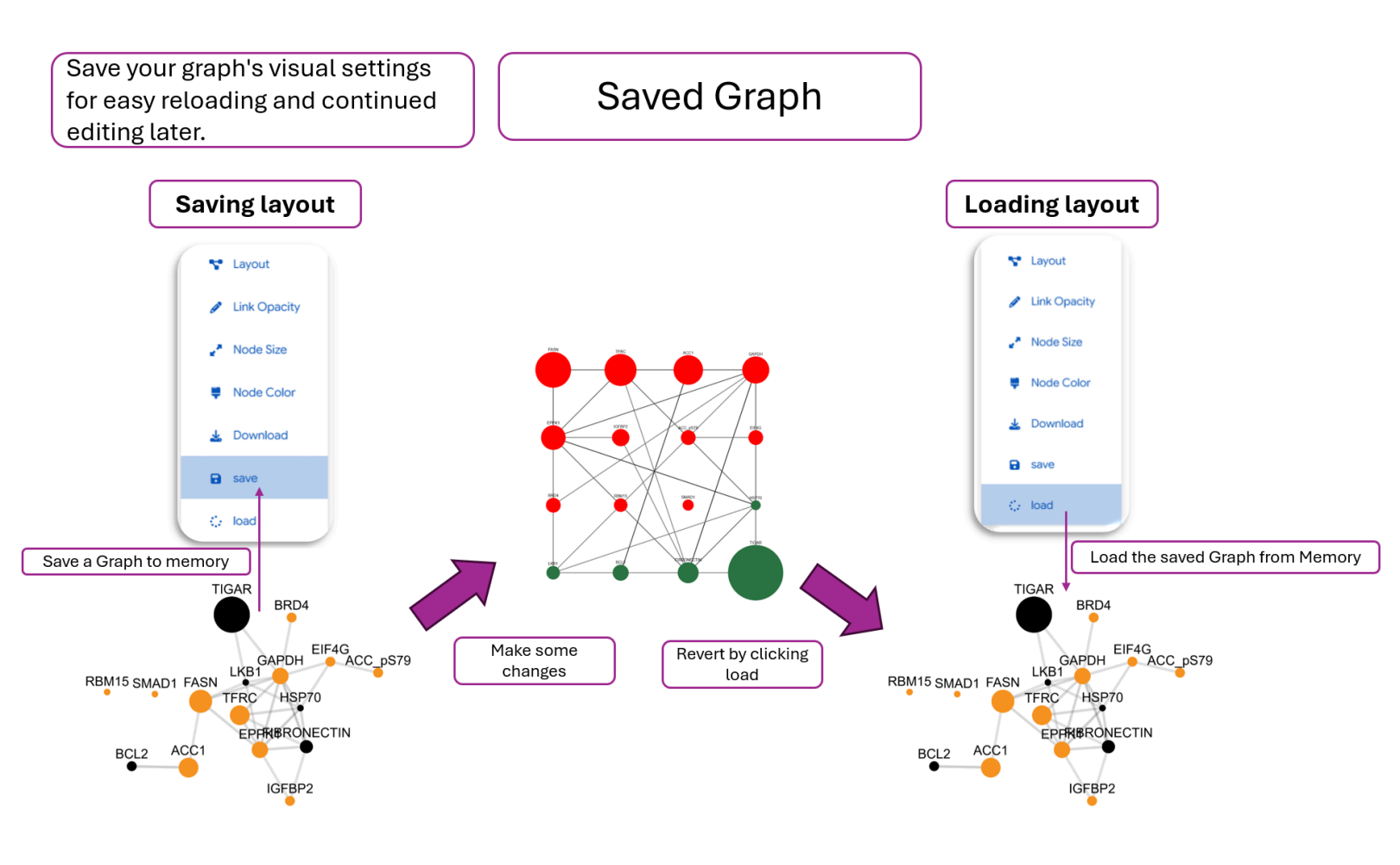
Adjust the node size. This helps change the ratio of node-text

Edge opacity controls the transparency of edges while maintaining their relative visibility.


Right Click Menu – edge's Opacity
LOWEST
LOW
NORMAL
MEDIUM
HIGH
HIGHEST

Adjust the visibility of edges to highlight stronger or weaker links.

Once the graph reaches the desired configuration, it can be saved using the Save option in the menu.


Saved Graph
Saving layout
Loading layout
TIGAR
BRD4
GAPDH
EIF4G
LKB1
ACC_pS79
RBM15
SMAD1
FASN
TFRC
HSP70
EPPK1
FIBRONECTIN
BCL2
ACC1
IGFBP2

Save your graph's visual settings for easy reloading and continued editing later.

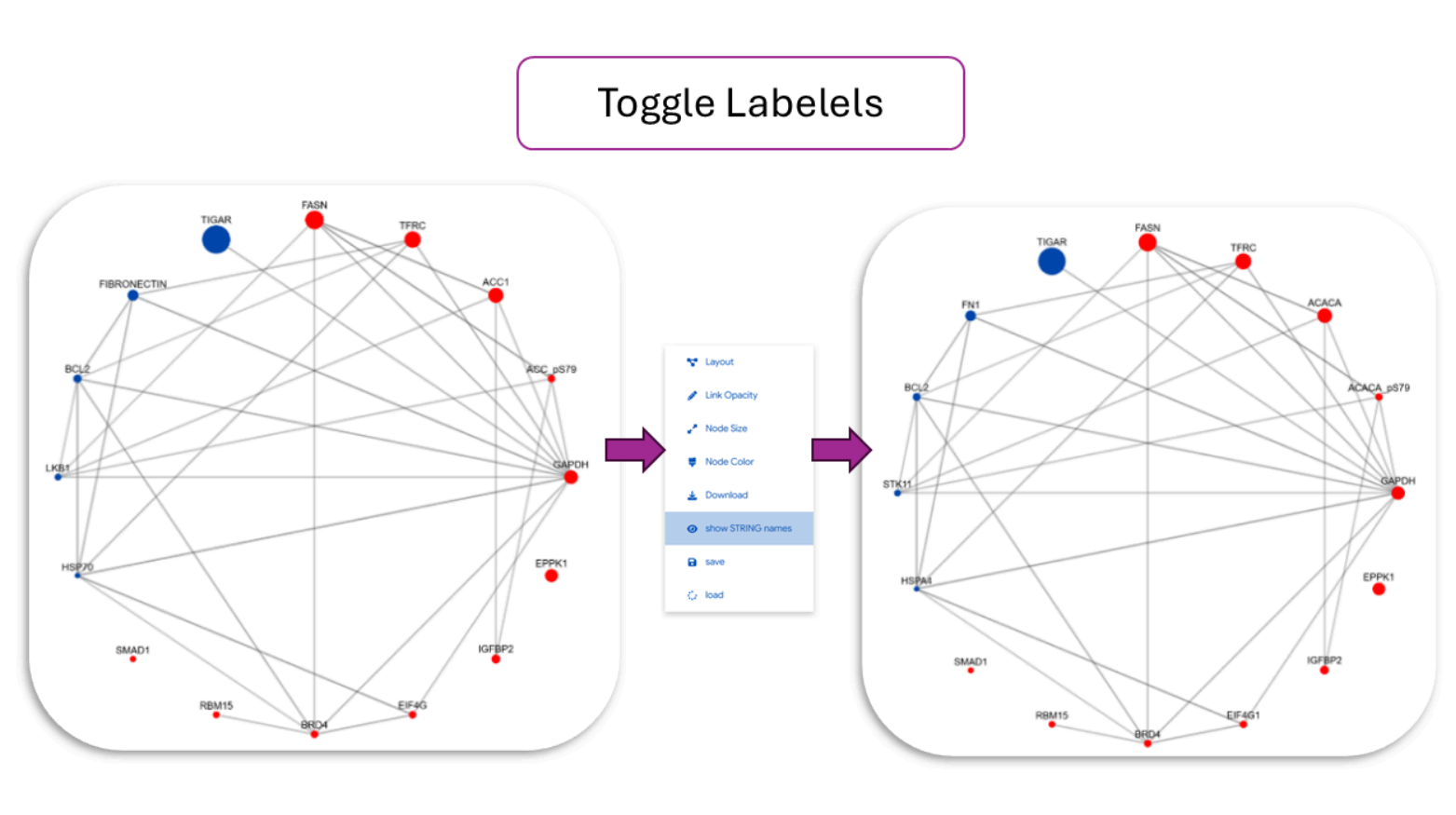
Save a Graph to memory

Make some changes

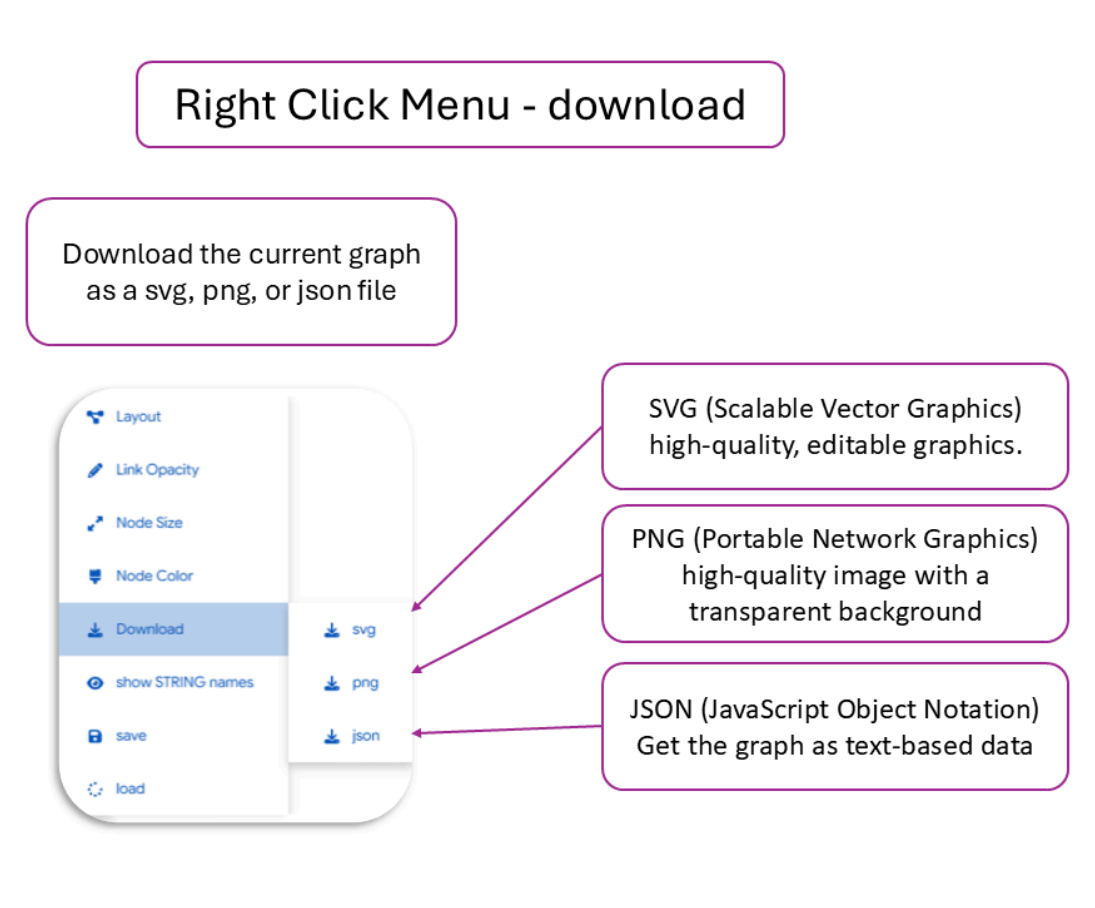
Revert by clicking load

Load the saved Graph from Memory

You can also toggle the labels and switch between original and STRING names.


Toggle Labelels

Alternativly, if you wish to continue working in another app, or export the result for further analysis, you can download the graph in one of 3 formats:
1. a .svg file (graph format)
2. a .png file (picture format)
3. a .json file (data format)


Right Click Menu - download
svg
png
json

Download the current graph as a svg, png, or json file

SVG (Scalable Vector Graphics) high-quality, editable graphics.

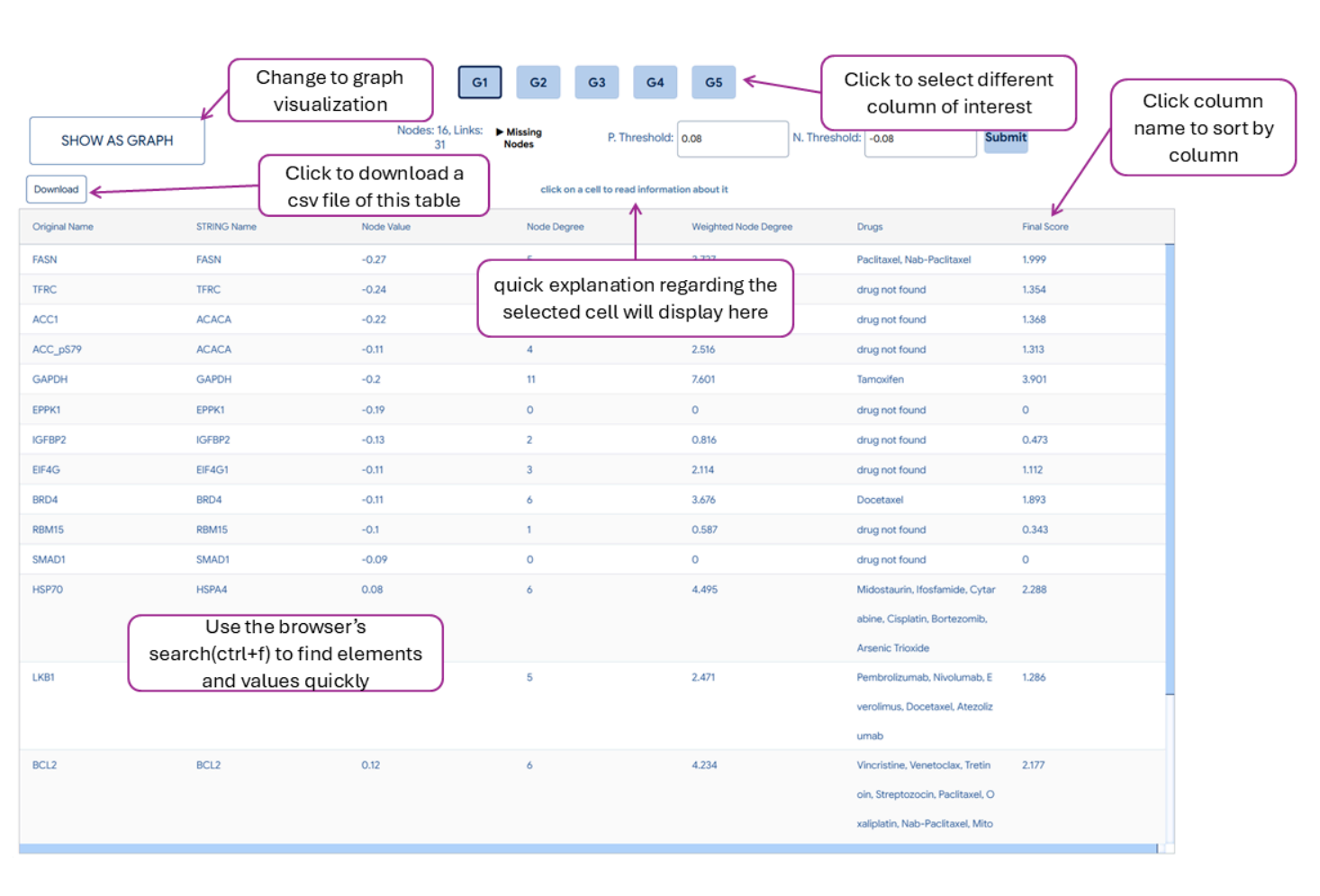
PNG (Portable Network Graphics) high-quality image with a transparent background

JSON (JavaScript Object Notation) Get the graph as text-based data

Table representation

The table view includes the following columns:

- Original name - the identifier provided in the input file
- STRING name - the matched identifier according to STRING
- Node value - the input score (e.g., derived from PCA or surprisal analysis)
- Node degree - number of interaction partners for each protein
- Weighted node degree - sum of weighted links based on STRING interaction probabilities
- Final score - average of the node size (absolute score value) and weighted node degree
- Drug - known anticancer drug retrieved from the Cancer Drugs Database


G1 G2 G3 G4 G5
SHOW AS GRAPH
Download
Nodes: 16, Links: 31
Missing Nodes
P. Threshold: 0.08
N. Threshold: -0.08
Submit
click on a cell to read information about it

Original Name	STRING Name	Node Value	Node Degree	Weighted Node Degree	Drugs	Final Score
FASN	FASN	-0.27	[illegible]	[illegible]	Paclitaxel, Nab-Paclitaxel	1.999
TFRC	TFRC	-0.24			drug not found	1.354
ACC1	ACACA	-0.22			drug not found	1.368
ACC_pS79	ACACA	-0.11	4	2.516	drug not found	1.313
GAPDH	GAPDH	-0.2	11	7.601	Tamoxifen	3.901
EPPK1	EPPK1	-0.19	0	0	drug not found	0
IGFBP2	IGFBP2	-0.13	2	0.816	drug not found	0.473
EIF4G	EIF4G1	-0.11	3	2.114	drug not found	1.112
BRD4	BRD4	-0.11	6	3.676	Docetaxel	1.893
RBM15	RBM15	-0.1	1	0.587	drug not found	0.343
SMAD1	SMAD1	-0.09	0	0	drug not found	0
HSP70	HSPA4	0.08	6	4.495	Midostaurin, Ifosfamide, Cytarabine, Cisplatin, Bortezomib, Arsenic Trioxide	2.288
LKB1			5	2.471	Pembrolizumab, Nivolumab, Everolimus, Docetaxel, Atezolizumab	1.286
BCL2	BCL2	0.12	6	4.234	Vincristine, Venetoclax, Tretinoin, Streptozocin, Paclitaxel, Oxaliplatin, Nab-Paclitaxel, Mito	2.177

Change to graph visualization

Click to select different column of interest

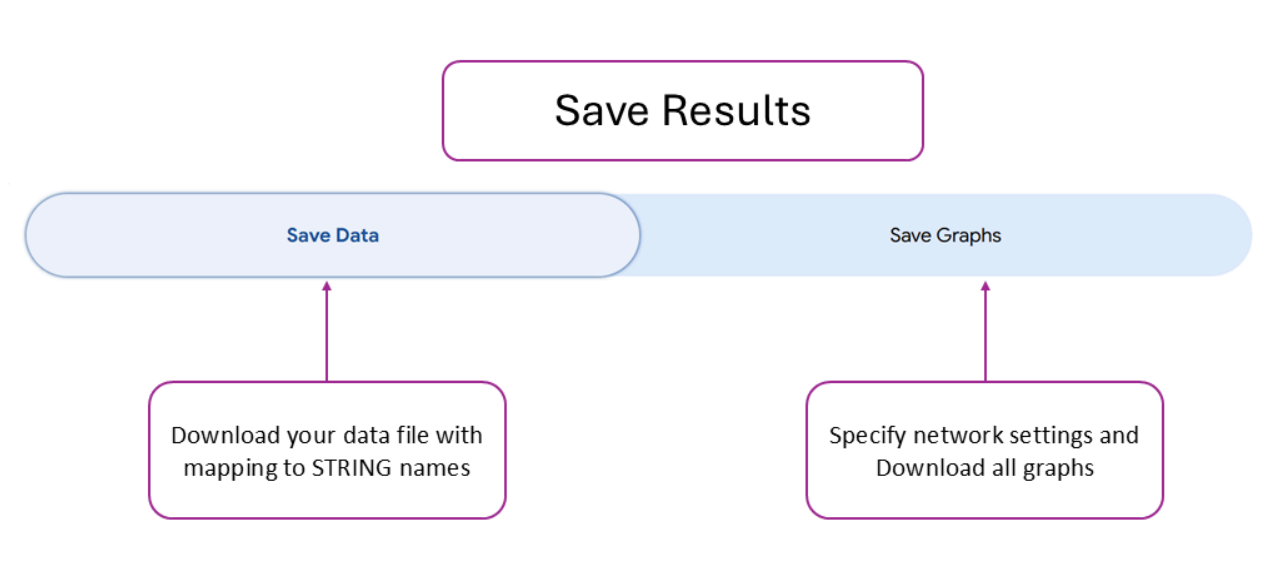
Click column name to sort by column

Click to download a csv file of this table

quick explanation regarding the selected cell will display here

Use the browser's search(ctrl+f) to find elements and values quickly

Users can save their work by clicking Save at the bottom of the Results window.


Save Results
Save Data
Save Graphs

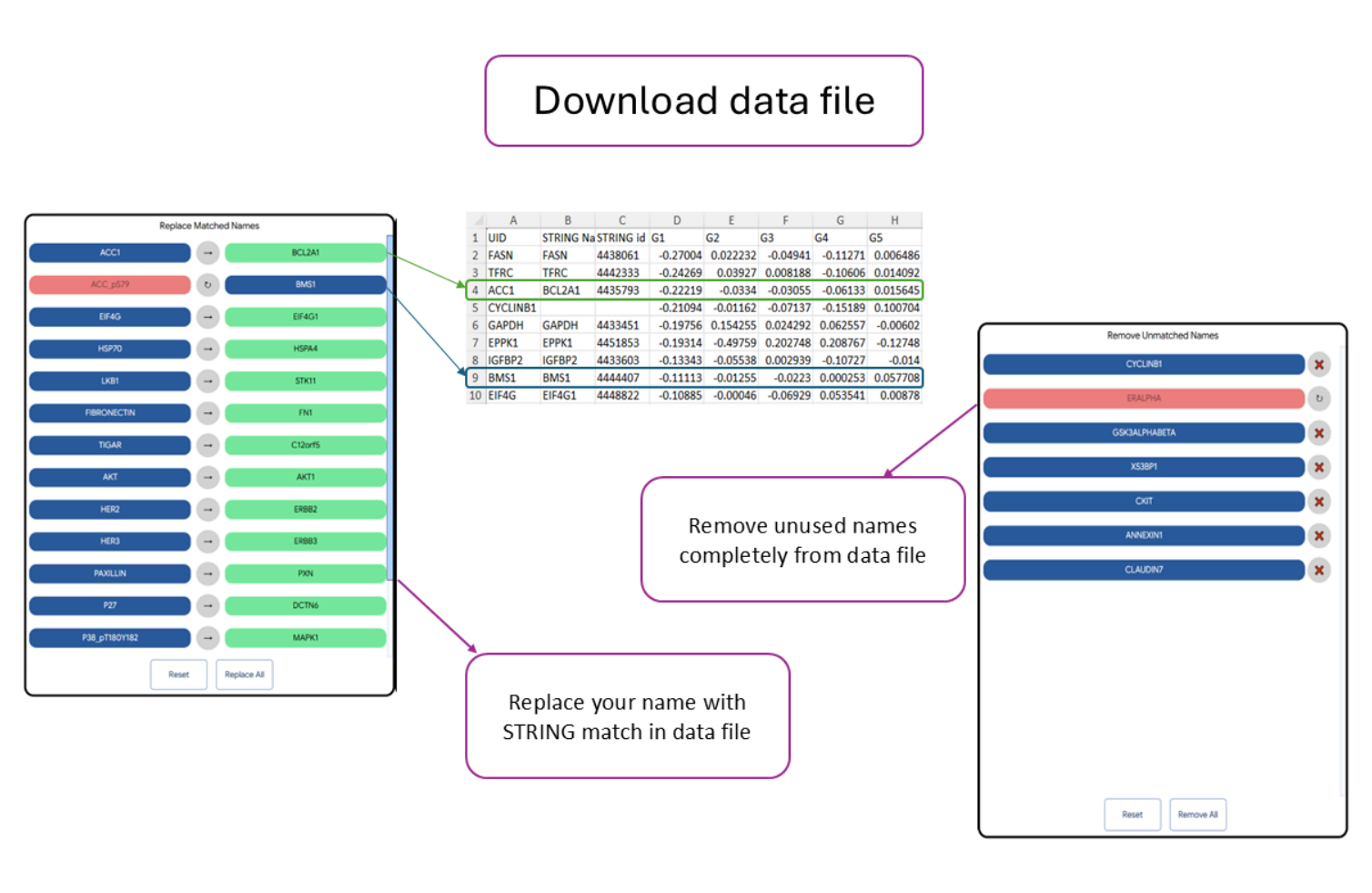
Download your data file with mapping to STRING names

Specify network settings and Download all graphs

Saving Data

Before downloading the modified dataset, users can review and adjust gene name mappings to STRING identifiers.
The left panel lists identifiers that have been matched or manually corrected. If desired, users can replace the original identifier with the STRING match using the arrow button.
The right panel lists identifiers that do not have a match in STRING.
Since these proteins do not appear in the network graphs, they may be removed from the dataset if desired.

The resulting xlsx file also contains 2 seperate sheets for the hyper-parameters used.
Note that it is unadvised to make changes to the downloaded file if you wish to upload it later to the platform for further analysis.


Download data file
Replace Matched Names
ACC1 BCL2A1
ACC_pS79 BMS1
EIF4G EIF4G1
HSP70 HSPA4
LKB1 STK11
FIBRONECTIN FN1
TIGAR C12orf5
AKT AKT1
HER2 ERBB2
HER3 ERBB3
PAXILLIN PXN
P27 DCTN6
P38_pT180Y182 MAPK1
Reset
Replace All
Remove Unmatched Names
CYCLINB1
ERALPHA
GSK3ALPHABETA
X53BP1
CKIT
ANNEXIN1
CLAUDIN7
Reset
Remove All

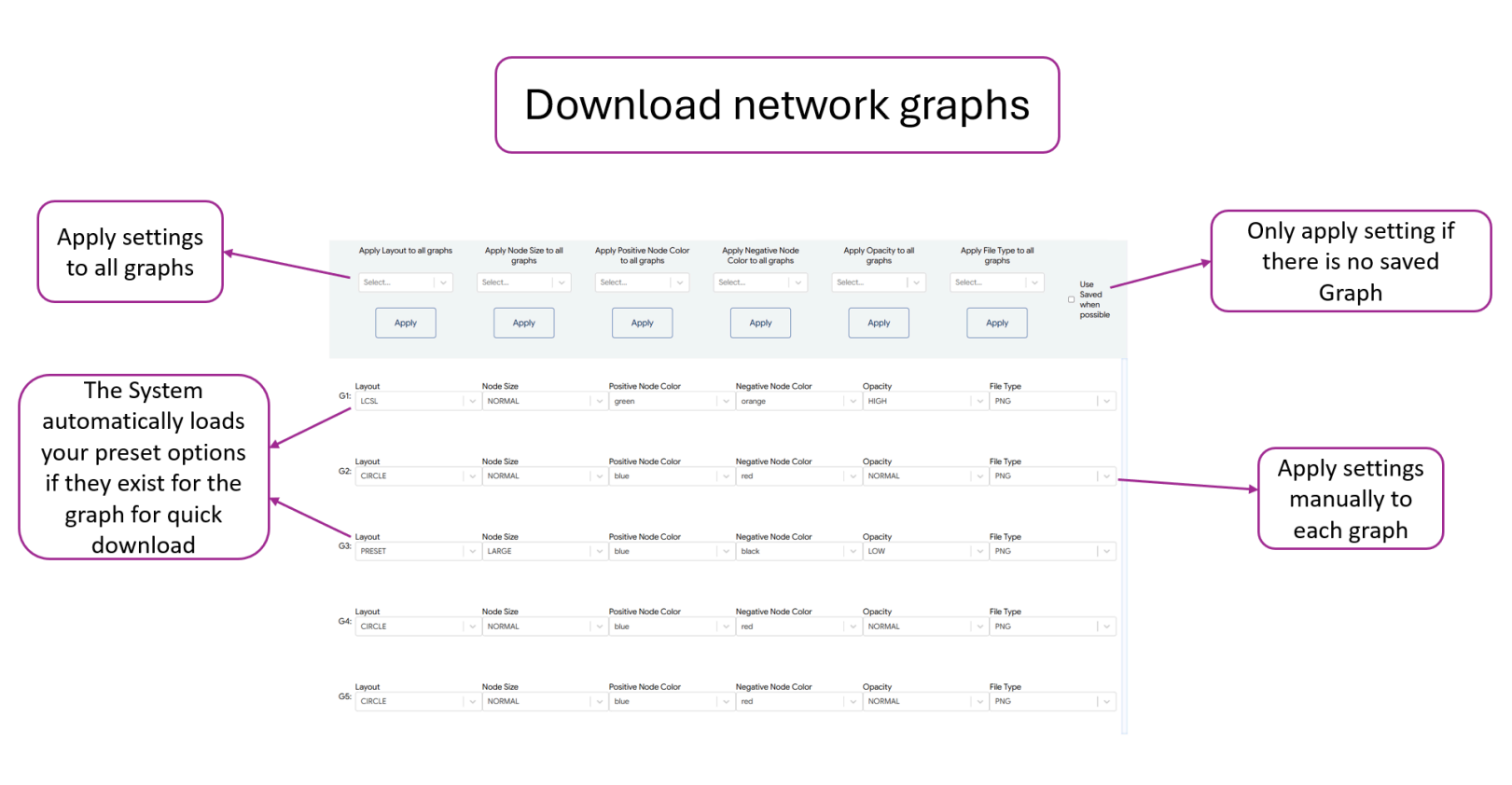
	A	B	C	D	E	F	G	H
1	UID	STRING Na	STRING Id	G1	G2	G3	G4	G5
2	FASN	FASN	4438061	-0.27004	0.022232	-0.04941	-0.11271	0.006486
3	TFRC	TFRC	4442333	-0.24269	0.03927	0.008188	-0.10606	0.014092
4	ACC1	BCL2A1	4435793	-0.22219	-0.0334	-0.03055	-0.06133	0.015645
5	CYCLINB1			-0.21094	-0.01162	-0.07137	-0.15189	0.100704
6	GAPDH	GAPDH	4433451	-0.19756	0.154255	0.024292	0.062557	-0.00602
7	EPPK1	EPPK1	4451853	-0.19314	-0.49759	0.202748	0.208767	-0.12748
8	IGFBP2	IGFBP2	4433603	-0.13343	-0.05538	0.002939	-0.10727	-0.014
9	BMS1	BMS1	4444407	-0.11113	-0.01255	-0.0223	0.000253	0.057708
10	EIF4G	EIF4G1	4448822	-0.10885	-0.00046	-0.06929	0.053541	0.00878

Remove unused names completely from data file

Replace your name with STRING match in data file

Saving Graphs

Users can also apply final adjustments before downloading the generated networks.
Global settings can be applied using the top menu (you must click apply for the settings to take effect).
Selecting “Use preset when available” applies saved layouts only to graphs that do not already have a stored layout when using the apply all menu.
If nodes were manually repositioned before saving, the custom layout will be preserved.


Download network graphs
Apply Layout to all graphs
Apply Node Size to all graphs
Apply Positive Node Color to all graphs
Apply Negative Node Color to all graphs
Apply Opacity to all graphs
Apply File Type to all graphs
Apply
Use Saved when possible

	Layout	Node Size	Positive Node Color	Negative Node Color	Opacity	File Type
G1:	LCSL	NORMAL	green	orange	HIGH	PNG
G2:	CIRCLE	NORMAL	blue	red	NORMAL	PNG
G3:	PRESET	LARGE	blue	black	LOW	PNG
G4:	CIRCLE	NORMAL	blue	red	NORMAL	PNG
G5:	CIRCLE	NORMAL	blue	red	NORMAL	PNG

Apply settings to all graphs

The System automatically loads your preset options if they exist for the graph for quick download

Only apply setting if there is no saved Graph

Apply settings manually to each graph

Browser compatibility

The following browsers and operating systems were tested for compatibility:

OS	Version	Chrome	Edge	Firefox	Safari
Windows	10/11	143	143	145	n/a
MacOS	Sonoma	143	143	n/a	17.3
Linux	PopOS	143	n/a	n/a	n/a